

I got the power

Principled Bayesian dynamic borrowing with power prior methods

Luiz Max Carvalho
lmax.fgv@gmail.com

Motivation

Paediatric RCT

PLUTO, $n = 92$

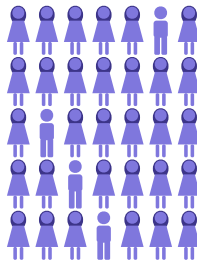


few, hard to recruit

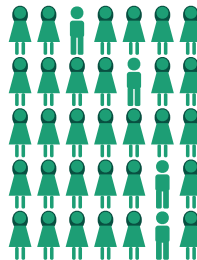
borrow?
← how much?

Adult trials

same drug, same endpoint



BLISS-52, $n = 548$



BLISS-76, $n = 577$

Let

- ⊙ $\mathbf{D}_0 = (D_0^{(1)}, \dots, D_0^{(n_0)}) \sim P_0 \equiv P_{\theta_0}$ be the **historical** data;
- ⊙ $\mathbf{D} = (D^{(1)}, \dots, D^{(n)}) \sim P_1 \equiv P_{\theta_1}$ be the **current** data.
- ⊙ Model is well-specified

Power prior

$$\pi_{\text{PP}}^{(\eta)}(\theta | D_0) := \frac{L(D_0 | \theta)^\eta \pi_0(\theta)}{c_0(\eta)},$$

Normalised power prior

$$\pi_{\text{NPP}}(\theta, \eta | D_0) := \pi_{\text{PP}}^{(\eta)}(\theta | D_0) \pi_A(\eta) \propto \frac{L(D_0 | \theta)^\eta}{c_0(\eta)} \pi_0(\theta) \pi_A(\eta).$$

$$\begin{aligned} p_{\text{NPP}}(\theta, \eta \mid D_0, D) &:= L(D \mid \theta) \pi_{\text{NPP}}(\theta, \eta \mid D_0), \\ &\propto \frac{L(D \mid \theta) L(D_0 \mid \theta)^\eta}{c_0(\eta)} \pi_0(\theta) \pi_A(\eta), \\ &\propto p(\theta \mid D) L(D_0 \mid \theta)^\eta \frac{\pi_A(\eta)}{c_0(\eta)}. \end{aligned}$$

- ⊙ $p_{\text{NPP}}^T(\theta \mid D_0, D) \equiv$ marginal effect; of central interest
- ⊙ $p_{\text{NPP}}^A(\eta \mid D_0, D) \equiv$ tells us something about compatibility

“Coherent” borrowing

How do we make $\eta \rightarrow 0$ when $P_0 \neq P_1$ and $\eta \rightarrow 1$ when $P_0 = P_1$?

Principled computation

NPP is doubly-intractable; can we compute exactly? Are there efficient **good** approximations?

Multiple historical data sets

Strange things (can) happen when you have multiple historical data sets.

Principled borrowing

Oliveira et al (2026+), [Shen et al. \(2026\)](#)¹:

Theorem (NPP shrinks to zero under discrepancy)

If if $P_0 \neq P_1$, under suitable conditions,

$$p_{\text{NPP}}^A(\eta \mid D_0, D) \rightarrow \delta_{\{0\}}(\eta),$$

$$a.s. P_{\theta_1}^n \otimes P_{\theta_0}^{n_0}.$$

Unwanted(?) discounting

However, when $P_0 = P_1$ the NPP does **not** concentrate at 1, and this is seen as a problem ([Pawel et al. , 2023](#)).

¹and others.

A tentative solution

Data-adaptive, fixed η

Choose η in a data-dependent manner, but keep it fixed

Information matching power prior (IMPP)

We choose η such that

$$\int_{\mathcal{X}^{k_n}} I_\eta(x) L(x \mid \theta_0) dx = \int_{\mathcal{X}^{k_n}} I_1(x) L(x \mid \theta_1) dx,$$

where $k_n = \lfloor (rn)^\nu \rfloor = \lfloor n_0^\nu \rfloor$ for $\nu \in [0, 1]$, $I_\eta(x) = \eta^2 \Delta(x)$, and $\Delta(x)$ is the Fisher relative information divergence of the posterior update from $p(\theta \mid D)$ to $p(\theta \mid D, x)$.

This is an adaptation of [Holmes & Walker \(2017\)](#).

IMPP and its properties

This setup leads to (Braga & Oliveira et al. (2026+)):

Information matching power prior (IMPP)

$$\hat{\eta}_{\text{IM}} = \left(\frac{\int_{\mathcal{X}^{k_n}} \Delta(x) L(x \mid \hat{\theta}_1) dx}{\int_{\mathcal{X}^{k_n}} \Delta(x) L(x \mid \hat{\theta}_0) dx} \right)^{1/2}.$$

Fixed variance normal example

Suppose that $y_{0i} \sim \mathcal{N}(\theta_0, \sigma^2)$ with likelihood $L_0(y_{0i} \mid \theta_0, \sigma^2)$, and $y_i \sim \mathcal{N}(\theta_1, \sigma^2)$.

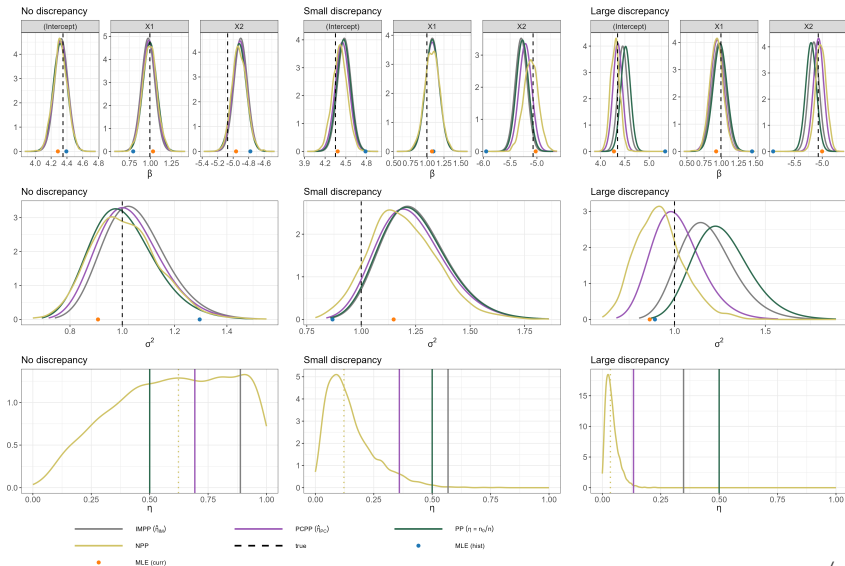
$$\eta = \left(\frac{\frac{v}{\sigma^2} + \frac{1}{\sigma^2}(m - \theta_1)^2 + \frac{1}{k_n}}{\frac{v}{\sigma^2} + \frac{1}{\sigma^2}(m - \theta_0)^2 + \frac{1}{k_n}} \right)^{1/2}.$$

Theorem

Let $D \sim P_{\theta_1}^n$ and $D_0 \sim P_{\theta_0}^{n_0}$ with $r = n_0/n$ a fixed constant. Under appropriate conditions, for a given sequence $\{k_n\}_{n=1}^\infty \subset \mathbb{N}$ with $k_n \leq n_0$ we have the following implications:

- (i) If $k_n \rightarrow \infty$, then $\hat{\eta}_{\text{IM}} = O_p(k_n^{-1/2})$ whenever $\theta_0 \neq \theta_1$, and, in particular $\hat{\eta}_{\text{IM}} \xrightarrow{p} 0$;*
- (ii) If $k_n/n \rightarrow 0$, then $\hat{\eta}_{\text{IM}} = 1 + O_p(k_n/n + n^{-1/2})$ whenever $\theta_0 = \theta_1$, and, in particular, $\hat{\eta}_{\text{IM}} \xrightarrow{p} 1$.*

Results



Variational Normalised Power Prior (VNPP)

- ⊙ Careful asymptotic analysis: we prove new and general BvM-type results for the PP and the NPP;
- ⊙ This motivates a MVN variational approximation $\rightarrow$ VNPP

Theorem

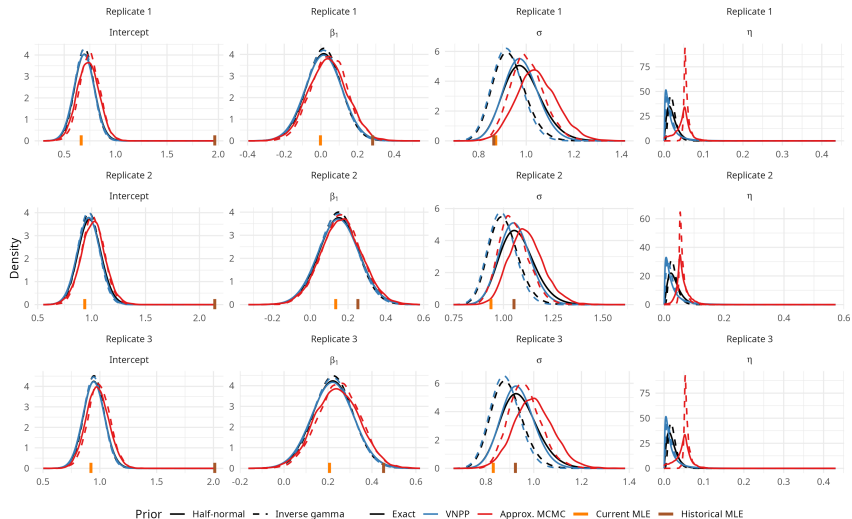
Suppose that $\theta_0 = \theta_1$. Under appropriate conditions, if $(1/n_0) \ll \varepsilon_{n_0} \ll 1$, we have that

$$\mathbb{P} \left[\int D_\eta(P_\theta, P_{\theta_0}) \tilde{p}_{\text{PP}}^{(\eta)}(d\theta \mid D, D_0) \leq c(r, \eta) \right] \geq 1 - \frac{1}{n_0 \varepsilon_{n_0}} h(r).$$

whence

$$c(r, \eta) := 2 \left(3 + \frac{1}{r} \right) \frac{1 + \eta}{1 - \eta} \varepsilon_{n_0},$$
$$h(r) := 1 + \frac{1}{r} + \frac{2}{3 + 1/r}.$$

VNPP works!



Multiple historical datasets are a menace

Normalisation matters

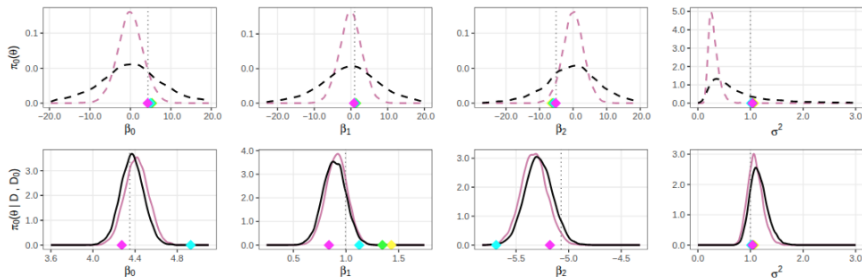
If you have data sets $D_{0k}, k = 1, 2, \dots, K$, how do you actually do things? Jointly or sequentially?

$$\pi_{\text{NPP}}(\boldsymbol{\theta}, \boldsymbol{\eta} \mid D_0) = \frac{\pi_A(\boldsymbol{\eta}) \pi_0(\boldsymbol{\theta}) \prod_{k=1}^K L_k(D_{0k} \mid \boldsymbol{\theta})^{\eta_k}}{c_0(\boldsymbol{\eta})},$$

$$\pi_{\text{NPP}_{\text{SEQ}}}(\boldsymbol{\theta}, \boldsymbol{\eta} \mid D_0) = \pi_A(\boldsymbol{\eta}) \prod_{k=1}^K p_{\eta_k}(\boldsymbol{\theta} \mid D_{0k}).$$

Consequences!

$\varepsilon = (1.5, 1.0, 0.5)$, Order: Compatible, Discrepancy: Small



Take home

It's possible to get "coherent" borrowing

But you give up on convergence speed

Variational approximations with guarantees

Beats the state-of-the-art in the accuracy-tractability trade-off

Multiple historical data sets

Still require a lot more research

Power priors provide semi-automatic learning

But other alternatives and connections (e.g. to hierarchical models) should be investigated – see [Shen et al. \(2026b\)](#)

THE
END